159302

Artificial Intelligence

Course and Assessment Guide

Computer Science
SMCS

S2 – 2023

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The course

Prescription

An advanced study of the principles and techniques used in developing Artificial Intelligence applications. The course includes the implementation and application of a range of AI methods including state-space representation and search, knowledge representation, constraint satisfaction, game playing, logic systems and machine learning.

Prerequisites: 159.201 or 159.234
Restrictions: 159.372

Learning outcomes

Students who successfully complete this course should be able to:

1. Describe the key Artificial Intelligence algorithms and problem-solving methods.
2. Design and implement computer programs to store and manipulate knowledge for AI applications.
3. Describe, and in some cases program, the special techniques that can be used to perform intelligent tasks such as searching, game playing, satisfying constraints, controlling, machine learning and natural language processing by computers.
4. Use logic for inference and fuzzy logic for real-time control problems.
5. Identify the advantages and disadvantages of various AI techniques for solving real-world problems

Textbooks *(Recommended reading)*

- Russell, S. and Norvig, P., Artificial Intelligence A Modern Approach (3rd edition). e.g. ISBN-13: 978-0-13-604259-4
- Neural Network and Fuzzy Logic Applications in C/C++ (Wiley Professional Computing) by Stephen Welstead
- Fuzzy Logic with Engineering Applications by Timothy Ross

Stream: Your online learning environment

Stream contains materials that can be downloaded to your private computer:

- Lecture slides, video lectures.
- Instructions for the assignments, relevant sample codes that can be used as a starting point for each assignment, and tips for testing your assignment solution.
- Tutorial questions to help test your level of understanding of the algorithms discussed.

Contacting your lecturer:

Auckland Campus: Dr Napoleon Reyes, email: n.h.reyes@massey.ac.nz
Manawatū Campus: Prof Hans Guesgen, email: h.w.guesgen@massey.ac.nz

How to approach your study

Each week covers a specific set of concepts, terminologies and algorithms organised according to the following main topics: Search, Constraint Satisfaction Problems, Games, Machine Learning, Fuzzy Logic, Logic and Natural Language Processing. You benefit if you do:

- Attend lectures / watch video lectures.
- Read the relevant lectures and reference materials that refer to the topic presented on that week.
- Solve the tutorial questions provided for the topic. Some algorithms may require some mathematical background; therefore, you should solve the problem sets that accompany the topics, to check and see if you misunderstood something.
- Complete the assignments on time. The lecture slides and start-up codes contain important clues that help with the completion of the assignments. Note that all assignment start-up codes were developed to run on a Windows 10 environment (as is the case with the machines at Massey's computer laboratories). Therefore, you should set-up a machine running the Windows 10 operating system, if you want to write programs at home, for your assignments.
- It is very important to code, compile, debug and run your own programs.

Suggested study schedule

This is a 15-credit course. The following table gives an idea of how to allocate your study time. Although the workload varies a little from week to week, you should allocate about 12.5 hours per week over the semester. If you are aiming for an A pass you may need to spend more time.

Reading the lectures slides and corresponding chapters from the recommended text.	36 hours
Assignments and Exercises	30 hours
Contact hours (lectures and tutorials)	36 hours
Assignments	36 hours
TOTAL	**138 hours**

Semester 1 2023 **Topics**

Date	Weeks	Topic
18 July	1	Course Overview Intelligent Agents, Demonstrations, Introduction to Search: Background and Motivation, Problem Solving Paradigm, Classes of Search, Cost and Performance, Any-Path Search Algorithms
25 July	2	Any Path Search Tutorial#1 Any Path Search Application: Robot Navigation - Tutorial#2
1 Aug	3	Optimal Search Algorithms: UC and A* (with or without the Expanded List)
8 Aug	4	Optimal Search – Tutorial#3 Optimality and Worst-Case Complexity, Optimal Search – Part 2 - Tutorial#4
15 Aug	5	**Assignment #1 Instructions, Start-up codes** Fuzzy Logic: Fuzzification, Inference Systems, Defuzzification, Fuzzy Logic Operators, Fuzzy Logic Engine Case Studies: Inverted Pendulum and Robot Navigation
22 Aug	6	Properties of Fuzzy Sets Fuzzy Logic - Tutorial#5 **Assignment #2 Instructions, Start-up codes**
28 Aug	MID SEMESTER BREAK	
4 Sept		
12 Sept	7	Machine Learning: Perceptron, Multi-layer Feed-Forward Networks Case Study: Letter Recognition Case Study: XOR Problem
19 Sept	8	Backpropagation Learning, Tutorial#6 Multi-output network, Tutorial#7
26 Sept	9	Constraint Satisfaction Problems (CSP): Motivation, Pure Backtracking (BT), Backtracking with Forward-Checking (BT-FC)

		CSP – Part 1 - Tutorial#8 CSP: Dynamic Variable Ordering, Incremental Repair, GSAT, WalkSat CSP – Part 2 -Tutorial#9
3 Oct	10	CSP (Other Strategies) CSP, Tutorial: Sensor Interpretation Games: Motivation, Board Games and Search, Minimax, Alpha-Beta Pruning. Case Study: IBM Deep Blue
10 Oct	11	Games - Tutorial#10 Logic: Motivation, Syntax, Semantics, Proof System, Sentences, Propositional Logic, Natural Deduction
17 Oct	12	Logic Proof System: Propositional Resolution Logic – Tutorial#11 Natural Language Processing: Technologies, Context-Free Grammar, Finite State Machine Parser Review for Finals
24 Oct	STUDY BREAK	
30 Oct	EXAMS	

Assessment

Assessment Description	**Learning Outcomes Assessed**					**Contribution to Course Marks**
	1	2	3	4	5	
Assignments		□	□	□	□	40.00%
Final Examination	□	□	□	□	□	60.00%

Assessment Schedule

Assessment	**Due Date / Deadline**	**Late Penalty**	**Course completion requirement**	**percentage**
Assignment 1	10/Sept. (tentative)	-10% per day	n/a	20
Assignment 2	10/October (tentative)	-10% per day	n/a	20
Final Exam	**(Exam timetable)**		**compulsory**	60

Electronic copies of previous exam courses are available online through Massey Library

https://www.massey.ac.nz/massey/research/library/help-and-instruction/how-to-find/examination-papers.cfm

Requirements for completing the course

To complete this course you will need to attend the Final Exam (compulsory) and submit the assignments. You also need to achieve an average value of at least 50% of the sum of the assessment items.

Assignment submission

Make sure that your assignments run on a Windows 10 machine before submitting. You can test them at Massey's computer laboratories for students. Assignments have to be submitted on the due date. If submitted late, there is a 10% per day penalty.

Academic integrity

It is mandatory that any assessment items that you submit during your University study are your own work. Massey University takes a firm stance on academic misconduct, such as plagiarism and any form of cheating.

Plagiarism is the copying or paraphrasing of another person's work, whether published or unpublished, without clearly acknowledging it. It includes copying the work of other students and reusing work previously submitted by yourself for another course.

Plagiarism, and cheating in tests and exams will be penalised; it is likely to lead to loss of marks for that item of assessment and may lead to an automatic failing grade for the course and/or exclusion from reenrolment at the University.

Please see the *Academic Integrity Guide for Students* on the University website for more information. The Guide steps you through the University Academic Integrity Policy and Procedures. For example, you will find definitions of academic integrity misconduct, such as plagiarism; how misconduct is determined and managed; and where to find resources and assistance to help develop the skills of academic writing, exam preparation and time management. These skills will help you approach university study with academic integrity.

Conditions for aegrotat pass and impaired performance

If you are prevented by illness, injury or serious crisis from attending a compulsory learning experience, an examination or completing an element of assessment (worth 10% or more) by the due date, or if you consider that your performance has been seriously impaired by such circumstances, you may apply for aegrotat or impaired performance consideration. You must apply on the Aegrotat & Impaired Performance Application form available from the Massey University website. The completed form must be accompanied by a certificate signed by a health professional, and/or corroborating evidence.

Queries about final grades

Each qualification is managed by a Programme Committee and the Chair of the Programme Committee, the Programme Director/Leader, is the chief examiner for that qualification. Although each Course Coordinator is the examiner for the course they coordinate, overall responsibility is with the Programme Director. The Programme Director/Leader ensures the processes have been followed, but accomplishes this through the examinations section of our Student Administration Unit. If you are not happy with the grade you are given, then you should, in the first instance, contact Student Administration about that. They will then ensure your complaint is followed up and a response provided.

Grievance procedures

A student who claims that he/she has sustained academic disadvantage as a result of the actions of a University staff member should use the University Grievance Procedures. Students, whenever practicable, should in the first instance approach the University staff member concerned. If the grievance is unresolved with the staff member concerned, the student should then contact the relevant Head of Institute/School/Department or College office for further information on the procedures. The procedures can be found on the University website in the University Calendar.